



Big Data Mining, digital tools, and methods to compare China and the West: The new agenda in global (economic) history

Manuel Perez-Garcia*

Shanghai Jiao Tong University, Department of History, School of Humanities, China / PI UMu BG22/00132 Project

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ABSTRACT

The paradigm of the new global (economic) history aims to present new case studies cross-referencing western and Asian sources and using new tools and methods from other disciplines such as digital humanities and computer sciences. Within such an interdisciplinary approach new digital solutions are being conceptualized and implemented to renew global history research and revise the Great Divergence debate. The GECM Project Database is presented as example that follows the paradigm of the “Complex-Systems Landscape” metaphor used and developed in this special issue. Thus, the core hypothesis in these papers is based on the role and agency of merchants and the networks they formed which ultimately changed consumer behavior and fostered the global circulation of goods in early modern China, Europe, and the Americas.

Introduction

This special issue presents a set of papers that reviews the last twenty years scholarship on global (economic) history and the Great Divergence debate which has overwhelmingly focused on large scale comparisons between Asia (mainly China) and Europe, frequently using macro-economic indicators (i.e., gross domestic product GDP). Such a line of research has presented dubious methodologies using conventional excel datasheets to apply GDP backwards projections from the present date to year 0. Essays in this special issue follow recent scholarship that has opened a new line of research applying Big Data to global (economic) history, storing new empirical evidence in computer databases, and using innovative quantitative methods which improve comparative research analysis of Asian and western economies. New case studies presenting new sources, methods, questions, and hypotheses are in line with the so-called new global (economic) history and so the articles of this special issue introduce the use of digital tools to analyze complex historical realities in regions of Asia, Europe, and the Americas.

In view of the latest global crises such as the Great Recession of 2008, the COVID-19 global pandemic, and the post-pandemic socioeconomic and political crisis, this special issue addresses thorough analysis and methods that can be applied to the understanding of those global turmoils and disruptions within an historical perspective and *longue durée* (long term) approach. Thus, within a context of profound revision and criticism, the articles of this special issue employ new empirical evidence and new quantitative methods to dig down beneath the historical foundations and causes of global historical processes and their effects on the

models of economic growth, modernization, and development among western and Asian regions. Political changes and transitions in the dawn of modern nation-states were preceded by global crises, which locally transformed economies and, in larger scale, the geopolitical scenario and balance of power, deepening the great divide and divergences between Asia, Europe, and the Americas. Likewise, such disturbances and causes have provoked potential convergences between such regions and shifts in the world economic order, international relations, and geopolitics.

Within this framework, this special issue aims to move beyond the Great Divergence debate by presenting a set of new case studies mining and analyzing new empirical evidence from Chinese and western historical archives to study the global circulation of goods, information, and technology, as well as the formation of social networks. The line of research within the new global (economic) history applying the Big Data Mining to early modern period, new quantitative methods, and digital tools, as well as using new databases to compare East Asia and western regions are the main issues covered. The designing and development of digital humanities systems to analyze complex historical realities should not be a goal in itself, but rather a method to find answers and clues to research questions and hypotheses using new historical sources within local, global, transnational, and comparative approaches.

The case studies of this volume aim to connect far-flung territories and economies within *longue durée* (long term) chronologies in the early modern period to analyze how global goods, information and trade networks had an effect on the economic development of East Asia (mainly China) and western regions. The analysis of complex historical systems

* Corresponding author at: Shanghai Jiao Tong University, School of Humanities, room 410C, 800 Dongchuan Road, Shanghai 200240, China.

E-mail addresses: mpargar@sjtu.edu.cn, manuel.perez@eui.eu

and social networks in Europe, Asia, and the Americas can be efficiently done using new databases such as the GECM Project Database and software such as GIS, GEPHI, UCINET or GENOPRO, among others. The current research articles give further explanation of the meaning and importance of employing such digital tools to implement social network analysis for the study of local and global spaces that comprehends socioeconomic, cultural, and political variables. Thus, the Great Divergence debate might be revised or potentially transformed into a new one to analyze the global dynamics that changed world economic systems, politics, and hegemonic powers by presenting new case studies within profound historical analysis and using historical sources and clear methodology.

Connections, comparisons, and polycentric approaches: the new agenda in global (economic) history

In the past two decades, the scholarship on global history has been seeking solutions and insights to the Great Divergence debate by connecting and comparing societies, economies, and politics across western and eastern regions. The goal was to study potential similarities and differences in the different paths of economic development and modernization between far-apart regions. A broad set of studies was carried out by the California School researchers (Vries, 2010) which moved away from and challenged Eurocentric or Anglo-centric approaches that had focused mainly on the British Empire as the driving power that led to the first industrial revolution. Research on Malthusian pressures, silver flows, institutions, markets, agricultural output, and standards of living, among other factors, forged the scholarship that sought to understand the positioning of East Asian economies in the early modern period and question European exceptionalism and supremacy (Duchesne, 2011; Flynn and Giraldez, 2010; Frank, 1998; Li, 1998; Pomeranz, 2000; von Glahn, 1996; Wong, 1997).

New theories, regional studies, and quantitative analysis were implemented, some of which proved to be valid, while others relied on the use of dubious economic macro aggregates (i.e., GDP) and have been sharply criticized (Allen, Bassino, Ma, Moll-Murata, and van Zanden, 2011; Broadberry, Guan, and Li, 2018; de Zwart and van Zanden, 2018; Madisson, 2007). The incremental research and use of quantitative methods that rely on such macroeconomic indicators have led global (economic) history and the Great Divergence debate to a stagnant position (O'Brien and Deng, 2017). In his timely latest book, Patrick O'Brien acknowledges this issue "after two decades of scholarly intellectual discourse among historians and social scientists with credentials and claims to the expertise required to engage in what has developed into the most famous example of the comparative approach to global history, discourse on the Great Divergence is now virtually over" (O'Brien, 2020).

The revision of the Great Divergence debate should be reformulated or amended, and, therefore, so should big questions such as why some economies grew faster than others or why, in the dawn of the first industrial revolution, northwestern European economies took off and East Asian (mainly China) economies stood behind. This aligns with the agenda of the so-called new global (economic) history, which stresses the idea that global history was somehow distorted by national narratives or the Cold War and post-Cold War discourse that perpetuated the great divide by seeking the study of hegemonic powers, firstly dominated by Eurocentric bias and lately by Sinocentric ones (Adelman, 2017; Perez-Garcia, 2021).

This new line of research relies upon interdisciplinary approaches within a geographic and polycentric basis, as well as avoidance of any sort of -ism (Eurocentrism, Sino-centrism, or American-centrism). There is not any dominant space either in Europe, Asia (China) or America in which world economic zones are equally important. The interconnection of these zones contributes to the intensification of links in terms of trade, circulation of goods, and technological and cultural transfers which intertwines local and globally socioeconomic, cultural, and politi-

cal trajectories. This clear-cut geographical application in global history based on a game of scales (micro and macro perspective) challenges national and patriotic narratives that overshadow and jeopardize the research agenda. This is the case of the scholarly obsession to seek the origins and date of globalization, which has been wrongly defined as component of global history.

Different research has insisted that globalization began from 1820 onwards through the relative global convergence of prices, with Britain being the engine of industrialization and modernization (Allen et al., 2011; de Zwart and van Zanden, 2018; O'Rourke and Williamson, 2004; Vries, 2015). Other studies date it to around 1571 with the opening of trade routes in the Pacific connecting colonial America with southern China via the well-known Manila-Acapulco galleon (Gruzinski, 2014; Martinez-Shaw, 2007; Yuste Lopez, 2007). Recent research affirms that globalization initiated circa 1519–1522 with the first world circumnavigation by Magellan and Elcano (Luque Azcona and Miranda Bonilla, 2020; Martinez-Shaw, 2019). In the case of China, the date used is around the 13th century when enterprises or historical events with globalizing features first occurred, such as the voyages of Marco Polo and the so-called Silk Road (Cheng, 2005; Wang, 2016). Thus, as Jeremy Adelman (2017) rightly mentions, there is a political bias and a noticeable national narrative permeated by an ethno-nationalism, whether Euro-centric, Sino-centric, or American-centric, which directly or indirectly can be observed in recent studies that seek to align themselves within global history.

A geographical definition without ideological biases that result from the current political agenda should be pursued, where globalization would begin when all heavily populated land masses began a sustained interaction, both directly and indirectly, that linked them deeply and permanently (Manning, 2020). In what could be defined as the "drama of hunger" (Perez-Garcia, 2018), the constant movement of tribes and ancient civilizations through migrations, diasporas, and exoduses in search of resources and means of subsistence, developed and intensified the links between continents and distant territories. Therefore, if we were to look for the starting date of the globalization process, it would begin in the Pleistocene when the continents ceased to be isolated and human movements began in Africa, Eurasia, America, and Australia.

The new global (economic) history components: Big Data Mining, digital tools, and new historical sources

The new global (economic) history endeavors lean on academic collaborations and use of new western and non-western primary sources, and, therefore, a training in eastern (Asian) languages. This should be accompanied by the digitization of historical sources (Drayton and Motadel, 2018). Thus, the renewal of global history and the revision of the Great Divergence debate, among others, require the implementation of new case studies to connect and compare western and eastern regions using and mining new empirical evidence, cross-referencing historical sources of diverse natures and archives, as well as presenting new digital solutions (new databases and softwares) to complex historical problems.

J.B. Owens's research aligns with the new global history methods to implement new digital solutions, conceptualization, and analysis of such complex realities of early modern economies and politics. Global history is a composite of complex systems and politics which can be modeled through the connection of its fragments. The analysis of historical events to reflect transformation and connection of world regions within a micro-perspective presents a challenge which researchers might solve by studying multiple socioeconomic, cultural, and political factors using new methodologies. Complex and non-linear human systems, connected trajectories and intertwined histories, can be accurately analyzed by separating features and systems' variables which are linked. The conceptualization of historical processes and events has been developed through the paradigm of social network analysis in which regional histories can provide answers and clues to grand questions related to modernization,

economic transformation, and transition from early modern to liberal regimes.

Owens presents the “Complex-Systems Landscape” metaphor or model (Keller, 1995, 2002) as a research solution to disentangle and understand world/global history through its components or microparts (de Vries, 2019) that shaped events, realities, and processes which locally effected global transformations. Such an approach aligns with the results of the GECM Project Database¹ (www.gecemdatabase.eu), partially presented in this special issue, which stands out as a new software solution to collect and analyze the new historical Big Data gathered in Chinese and European archives (Perez-Garcia and Diaz-Ordoñez, 2023). Some historians dealing with digital humanities mistakenly present creation of databases as a goal in itself through voluminous and vague repositories of data. GECM Project Database is conceived and implemented as a tool with the aim to develop the GECM Project’s main hypothesis, which sustains that merchants’ intermediation changed patterns of consumption through the circulation of global goods in both China and western regions. Merchants played a crucial role in creating long distance networks and informal institutions that bypassed state supervision.

Therefore, new empirical, archival evidence has been cross-referenced within the GECM Project Database. Such a methodological tool integrates digital humanities and global history with new historical Big Data in a new database management system (DBMS) enabling the effective implementation of computational and quantitative methods for the analysis of information. Big Data as applied to global (economic) history is based on clustering information and cross-referencing historical sources of diverse natures, which can be stored and accessed according to current computing and quantitative methods. This methodological approach that supports new research, case studies, and analysis is exemplified in the papers presented in this special issue. Thus, a new generation of global historians seeks to identify comparisons and connections across regions for the early modern period through utilization of new sources to quantify the circulation of global goods, social networks, and local economies.

Under this framework GECM Project Database is Open Access, selecting a MariaDB architecture for the core software of the database. This management system, derived from the traditional MySQL, has a General Public License (GPL), and its drivers of high compatibility with SQL standards ensure future modifications. Presently, the GECM Project Database incorporates nearly 40,000 registers of Chinese, European and American goods which were introduced in East Asia and western markets. Various sources from historical archives in China and Europe such as the Historical Archive of Macao, First Historical Archives of China, *Archivo General de Indias* in Seville, and *Archive de la Chambre de Commerce de Marseille* are stored in the GECM Project Database. This digital solution facilitates cross-referencing historical sources of diverse natures such as local gazetteers [*zhōngguó dìfāng zhì* 中国地方志], trade records, customs duties, and probate inventories, among others. The database is built within a set of four general queries: 1) analysis of fashions over time; 2) Asia-Europe trade relations; 3) exports/intermediation of Asian goods and consumption; and 4) consumption, social status, and residence.

The “Complex-Systems Landscape” metaphor in Owen’s paper can be applied to the analysis of different systems and politics across time and space. Likewise, the GECM Project Database is conceptualized as a digital solution and tool to empirically analyze the evolution, transformation, and connection of distant territories through the circulation of global goods and formation of social networks. A good example is the spatial organization and transmission of information through maritime

routes that connected territories of the Spanish empire. Both overland and maritime circuits of mail and posts main in regions of South America such as the viceroyalty of New Granada through the Cartagena de Indias - Santa Fe de Bogotá route stands out as a good case study, presented by Moreno-Cabanillas, for the analysis of information management and negotiations between social actors in the peripheries of the Spanish Empire.

Information was crucial to obtain the know-how to operate in unregulated markets when the action and presence of the state seemed inefficient to control the imports of overseas goods from China. Thus, New Granada routes and networks of information, depicted in Moreno-Cabanillas’s paper, are complemented by the analysis and trade routes of New Spain’s markets presented in Ibarra’s paper. The mediation, agency and negotiations of social actors can be studied through the implementation of relational analysis using GEPHI or UCINET software. This allows us to visualize through the graphs and nodes of social network analysis the mutation of the networks and actors’ strategies in terms of access to information; efficiency of state institutions; and the structure of informal institutions, alliances, and competitions in local markets that effected global trade and circulation of Chinese goods in American and European regions. The identification of egocentric or eccentric networks which measure the distance between the key social actors helps us to determine the formation of monopolies and accumulation of power by local elites and families.

Ibarra’s analysis of trade companies, fiscal records and probate inventories unveils merchants’ interests, business practices and strategies based on family or non-family ties. Thus, the construction of spatial relations and social networks is a paramount tool, based on complex realities solutions using GECM Project Database, to identify trade circuits of Chinese goods and the role of traders in introducing such goods in local markets (Perez-Garcia, Wang, Svriz-Wucherer, Fernández-de-Pinedo, and Diaz-Ordoñez, 2022). A good example is the position and strategies mobilized by communities such as the Jesuits, who acted not only as religious agents, but mainly as merchants as they formed business and financial networks that connected European regions with the Americas and Asia during the early modern period. Incremental scholarship has mainly focused on the evangelical and missionary function of the Society of Jesus. Svriz’s paper connects the frontiers of empires in the Pacific, with the Philippines as main node, to analyze the role and agency of the Jesuits in the circulation of information and trade strategies to finance global trade between the Chinese and Iberian empires.

The comparison of two Jesuit missions and trade entrepôts in Macau and Paraguay is a novel case study to analyze global trade routes and goods that circulated trans-continently and affected local markets, consumer behavior, and cultural habits. The craze for Chinese silks in western regions was fostered by the formation of these social networks, which introduced in western markets a wide array of silk typologies produced in the local markets of China’s Jiangnan region. Wang Li’s paper presents an analysis on the interest in consuming silks of different qualities and imported fabrics and manufactures by different social groups during the Qing dynasty and how this affected cultural habits and changed ordinary life.

Conclusions

Once the Great Divergence debate has been overcome and revised based on the premises of the new global (economic) history and the implementation of new case studies, this special issue has shown this new line of research and composites that rely upon Big Data Mining, new digital tools, and cross-referencing historical sources of diverse nature. The “Complex-Systems Landscape” metaphor or model presented in this research contributes to the understanding of global history’s intertwining realities and events in which clues and answers to big questions can be found in local economies and transformations. The GECM Project Database, used in the papers of this special issue as a new digital solu-

¹ GECM Project Database is authored by Perez-Garcia and Diaz-Ordoñez (2021). GECM Project Database version 2021. The Database is published in Open Access: www.gecemdatabase.eu/index.html. www.gecemdatabase.eu.

tion to analyze circulation of global goods and trade, precisely follows the paradigm of the “Complex-Systems Landscape” metaphor applied by J.B. Owens. The aim of GECEM Project Database is to analyze changes in patterns of consumption, global circulation of goods, and the formation of trade networks in local markets of China and Europe. This is arduous but exciting research, since using new digital solutions, trade between regions and localities in Asia, America and Europe can be analyzed. Furthermore, implementing new quantitative methods to mine new data facilitates the understanding of potential divergences and convergences among long-distance economies. Therefore, it is possible to establish a global map, with local variables, of the myriad trade networks, the mutation of the networks themselves (using GEPHI software), and the family coalitions and marriage strategies that are visualized in the GECEM Project Database using the GENOPRO genealogy software. This research thus takes on an innovative and cutting-edge dimension that renews the global (economic) history in China and Europe.

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